Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 2024-2025

Comprehensive Examination
(EC-3 Regular)

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Open Book
Weightage : 50%
Duration : 2 Hours
Date of Exam : 29-03-2025_FN

No. of Pages = 9
No. of Questions = 5

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

1 a) BITS Pilani University researcher wants to determine whether gender (Male or Female) is independent of preference for a new product (Like or Dislike). A random sample of 250 people is surveyed, and the results are recorded in the contingency table below: [4 Marks]

Gender	Like	Dislike	Total
Male	80	40	120
Female	90	40	130
Total	170	80	250

Test at a 5% significance level whether gender and preference for the product are independent.

Solution:
Step 1: Define the Hypotheses
H0: Gender and preference for the new product are **independent**.

H1: Gender and preference for the new product are **not independent**.------- 1M

Step 2: Compute Expected Frequencies
The expected frequency (E) for each cell in a contingency table is calculated as:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

For each cell:

$$E_{Male,Like} = \frac{120 \times 170}{250} = \frac{20400}{250} = 81.6$$

$$E_{Male,Dislike} = \frac{120 \times 80}{250} = \frac{9600}{250} = 38.4$$

$$E_{Female,Like} = \frac{130 \times 170}{250} = \frac{22100}{250} = 88.4$$

$$E_{Female,Dislike} = \frac{130 \times 80}{250} = \frac{10400}{250} = 41.6$$

Gender	Like (Observed)	Like (Expected)	Dislike (Observed)	Dislike (Expected)
Male	80	81.6	40	38.4
Female	90	88.4	40	41.6

------- 1M

Step 3: Compute the Chi-square Test Statistic

The formula for **Chi-square** is:

$$\chi 2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(80-81.6)^2}{81.6} + \frac{(40-38.4)^2}{38.4} + \frac{(90-88.4)^2}{88.4} + \frac{(40-41.6)^2}{41.6}$$

$$= \frac{(-1.6)^2}{81.6} + \frac{(1.6)^2}{38.4} + \frac{(1.6)^2}{88.4} + \frac{(-1.6)^2}{41.6}$$

$$= \frac{2.56}{81.6} + \frac{2.56}{38.4} + \frac{2.56}{88.4} + \frac{2.56}{41.6}$$ ------- 1M

$$= 0.0314 + 0.0667 + 0.0290 + 0.0615 = 0.1886$$

Step 4: Find the Critical Value

Degrees of freedom (df) = **(rows - 1) × (columns - 1) = (2 - 1) × (2 - 1) = 1**

Significance level (α) = **0.05**

From the Chi-square table, the critical value for df=1, at alpha = 0.05 is **3.841**.

Step 5: Compare & Make a Decision

$\boldsymbol{\chi^2 = 0.1886 \text{ is less than } 3.841, \text{ we fail to reject H0.}}$

Step 6: Conclusion

At the **5% significance level**, there is **no significant evidence** to suggest that gender and preference for the new product are related. This means gender and product preference **are independent**. ------- 1M

b) A machine learning instructor claims that the average time students spend on a deep learning project is 80 hours**.** A group of data science students believes that the actual time is different. To test this claim, a sample of **15 students** was taken, and their recorded project completion times (in hours) were:

76, 85, 90, 78, 83, 88, 91, 75, 84, 79, 87, 92, 80, 86, 89

Assume the time spent follows a normal distribution**,** but the population standard deviation is unknown**.** At 3% significance level**,** can the students conclude that the actual mean project completion time differs from 80 hours? [4 Marks]

Solution:

Sample Mean ($\bar{x}$) = 84.2

Sample Standard Deviation (s) = 5.52

$H_0: \mu = 80$, $H_1: \mu \neq 80$ ------- 1M

$t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$ = 2.95 (P-value: 0.0106) ------- 1.5M

α=0.03, critical t-value for $t_{0.015, 14}$ = 2.624 ------- 0.5M

The critical for 0.015 **in the upper tail** is **2.625.** (Two-tailed test)

Since **2.625 < 2.95, Reject** H0.

Conclusion: There is sufficient evidence at the 3% significance level to conclude that the actual mean project completion time differs from 80 hours. ------- 1M

2 a) A Professor wants to build a prediction model based on the given data of height (x) and the weight(y) of each student. Build a model to predict:
"The height, based on the weight of the student" [4 Marks]

x	80	69	73	76	72	77	59	76	67	58
y	86	112	114	96	57	79	94	74	106	76

Solution:

Height (x)	Weight (y)	x^2	y^2	x*y
80	86	6400	7396	6880
69	112	4761	12544	7728
73	114	5329	12996	8322
76	96	5776	9216	7296
72	57	5184	3249	4104
77	79	5929	6241	6083
59	94	3481	8836	5546
76	74	5776	5476	5624
67	106	4489	11236	7102
58	76	3364	5776	4408
707.00	**894.00**	**50,489.00**	**82,966.00**	**63,093.00**
70.70	**89.40**	**5,048.90**	**8,296.60**	**6,309.30**

Table calculations (2.5M)

$$bxy = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum y^2 - (\sum y)^2}$$

$$= \frac{10 \cdot 63093 - 707 \cdot 894}{10 \cdot 82966 - (894)^2}$$

$$= \frac{630930 - 632058}{829660 - 799236}$$

$$= -\frac{1128}{30424}$$

$$= -0.0371$$

Regression Line x on y

$$x - \bar{x} = bxy\left(y - \bar{y}\right)$$

$$x - 70.7 = -0.0371(y - 89.4)$$

$$x - 70.7 = -0.0371y + 3.3146$$

$$x = -0.0371y + 3.3146 + 70.7$$

$$x = -0.0371y + 74.0146$$

Calculation for finding regression coefficient and line (1.5M)

2 b) In a particular city, the number of car accidents per day follows Poisson distribution. On five randomly chosen days, the number of accidents recorded was 4, 2, 5, 3, and 6.
i) Find the maximum likelihood estimate (MLE) of the average number of accidents per day.
ii) Using the MLE from part (i), estimate the probability that there will be no accidents on a randomly chosen day. [4 marks]

Solution:

Suppose we have data generated from a Poisson distribution. We want to estimate the parameter of the distribution

The probability of observing a particular random variable is $P(X;\mu)=\frac{e^{-\mu}\mu^{X}}{X!}$

$$P(X_1,X_2,\ldots,X_n;\mu)=\frac{e^{-\mu}\mu^{X_1}}{X_1!}\times\frac{e^{-\mu}\mu^{X_2}}{X_2!}\times\ldots\times\frac{e^{-\mu}\mu^{X_n}}{X_n!}$$

$$L(\mu;\mathbf{X})=\prod_i e^{-\mu}\mu^{X_i}$$

$$L(\mu;\mathbf{X})=e^{-n\mu}\mu^{n\overline{X}}$$

$$L(\mu;\mathbf{X})=e^{-n\mu}\mu^{n\overline{X}}$$

$$\ell(\mu;\mathbf{X})=-n\mu+n\overline{X}\log\mu$$

$$\frac{d\ell}{d\mu}=-n+\frac{n\overline{X}}{\mu}=0$$

$$\hat{\mu}=\overline{X}$$

Finding maximum likelihood function (2.5M)

The data for the number of accidents on the 5 randomly chosen days is: 4, 2, 5, 3, and 6.

$$\lambda_{\text{MLE}}=\frac{1}{5}\sum_{i=1}^{5}x_i=\frac{1}{5}(4+2+5+3+6)=\frac{20}{5}=4$$

Thus, the **maximum likelihood estimate (MLE) of the average number of accidents per day is** $\lambda=4$.

For a Poisson distribution, the probability of observing exactly k accidents on a given day is given by the formula:

$$P(X=k)=\frac{\lambda^k e^{-\lambda}}{k!}$$

calculations (1.5M)

$$P(X=0)=e^{-4}\approx 0.0183$$

Thus, the estimated probability that there will be no accidents on a randomly chosen day is approximately **0.0183**, or about **1.83%**.

3 a) A retail chain wants to forecast monthly sales for a popular product using a 3-month moving average method. The sales figures for the past six months (in thousands of units) are: [4 marks]

Month	Sales (in '000 units)	Month	Sales (in '000 units)
January	50	April	65
February	55	May	70
March	60	June	80

i) Compute the 3-month moving average for April, May, and June.

ii) Predict the sales for July using the moving average method.

Solution: (i)

- **3-month moving average for April** = **55** (thousands of units)
- **3-month moving average for May** = **60** (thousands of units)
- **3-month moving average for June** = **65** (thousands of units) -------- 2M

(ii)

$$\text{Moving Average for July} = \frac{65+70+80}{3} = \frac{215}{3} \approx 71.67$$

The predicted sales for **July** using the 3-month moving average method is approximately **71.67 thousand units**.

---2M

b) A time series X_t is given as follows: [4 Marks]

$X_t = \{4, -2, 3, -1, 5, -3\}$

i) Compute the sample **autocorrelation** at lag **1** (ρ1).

ii) Based on the result, determine if the process is likely to be white noise.

Solution: (i)

The **sample autocorrelation at lag** k is defined as:

$$\rho_k = \frac{\sum_{t=1}^{n-k}(X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^{n}(X_t - \bar{X})^2}$$

Where:

- X_t is the value of the time series at time t,
- $\bar{X}$ is the sample mean of the time series,
- n is the total number of observations,

The sample mean is:

$$\bar{X} = \frac{1}{6}(4 + (-2) + 3 + (-1) + 5 + (-3)) = \frac{6}{6} = 1$$

So, the sample mean $\bar{X} = 1$.

We now compute this sum for each pair of observations X_t and X_{t+1}:

$$(X_1 - \bar{X})(X_2 - \bar{X}) = (4-1)(-2-1) = 3 \times (-3) = -9$$

$$(X_2 - \bar{X})(X_3 - \bar{X}) = (-2-1)(3-1) = (-3) \times 2 = -6$$

$$(X_3 - \bar{X})(X_4 - \bar{X}) = (3-1)(-1-1) = 2 \times (-2) = -4$$

$$(X_4 - \bar{X})(X_5 - \bar{X}) = (-1-1)(5-1) = (-2) \times 4 = -8$$

$$(X_5 - \bar{X})(X_6 - \bar{X}) = (5-1)(-3-1) = 4 \times (-4) = -16$$

Now sum these values:

$$-9 + (-6) + (-4) + (-8) + (-16) = -43$$

Similarly $\sum_{t=1}^{n}(X_t - \bar{X})^2$ = 58.

$$\rho_1 = \frac{-43}{58} \approx -0.741$$

So, the **sample autocorrelation at lag 1** (ρ_1) is approximately **-0.741**. ------- 3M

(ii) Since the autocorrelation at lag 1 is far from zero, the process is **not likely to be white noise**. A white noise process would typically have autocorrelations near zero, indicating that the values are independent and uncorrelated over time. --------------1M

4) A researcher wants to study the effect of three different diets on weight loss. They randomly assign 15 individuals to three diet groups (Diet A, Diet B, and Diet C), with 5 participants in each group. After 8 weeks, the recorded weight loss (in kg) for each participant is as follows:

Diet A	3.2	2.8	3.5	3.1	2.9
Diet B	4.1	3.9	4.3	4.0	4.2
Diet C	2.5	2.7	2.8	2.6	2.9

Perform a one-way ANOVA test to determine if there is a significant difference in mean weight loss among the three diet groups at 5% level of significance. [8 marks]

Solution:

H_0 : There is no difference in mean weight loss among three diet groups.

i.e. $\mu_A = \mu_B = \mu_C$. 1 mark

H_1 : There is a difference in mean weight loss among groups.

i.e. $\mu_A \neq \mu_B \neq \mu_C$ 1 mark

						Total (C_i)
Diet A :	3.2	2.8	3.5	3.1	2.9	15.5
Diet B :	4.1	3.9	4.3	4.0	4.2	20.5
Diet C :	2.5	2.7	2.8	2.6	2.9	13.5
				Grand Total (G =)		49.5

1 mark

mean of all data = $\bar{x} = \frac{49.5}{15} = 3.3$

$\bar{x}_1 = \frac{C_1}{5} = 3.1$, $\bar{x}_2 = \frac{C_2}{5} = 4.1$, $\bar{x}_3 = \frac{C_3}{5} = 2.7$ 1 mark

ANOVA Table:

Source of variation	d o f	Sum of Sq.	Mean Sum of Sq.	F-ratio
Between groups	$k-1=2$	$SSTR = 5.2$	$MSTR = \frac{SSTR}{k-1} = 2.6$	$F = \frac{MSTR}{MSE}$ 62.3500
within groups	$n-k=12$	$SSE = 0.5$	$MSE = \frac{SSE}{n-k}$ 0.047	
Total		$SST = 5.7$		

2 mark

For $\alpha = 0.05$, F_{tab} with $(2,12)$ dof is 3.89

1 mark

$\because$ $F_{tab} < F_{cal}$ i.e. $3.89 < 10.4$ } hence reject H_0 or Accept H_1

Hence there is a significant difference b/w mean weight loss among groups

1 mark

5 a). Find the exponential smoothing for α=0.4 and α=0.6 for the following data and find out which weighting factor gives better smoothing. [5 Marks]

Month/ Year	Deflection	Month/ Year	Deflection
1/2024	-213	7/2024	-420
2/2024	-564	8/2024	-360
3/2024	-35	9/2024	203
4/2024	-15	10/2024	-338
5/2024	141	11/2024	-431
6/2024	115	12/2024	194

Solution: For alpha = 0.4

Month	Smoothed Value (S_t)
1/2024	−213
2/2024	−353.4
3/2024	−226.04
4/2024	−141.624
5/2024	−28.5744
6/2024	28.85536
7/2024	−150.686784
8/2024	−234.411
9/2024	−59.4466
10/2024	−170.868
11/2024	−274.9208
12/2024	−87.35248

For alpha= 0.6

Month	Smoothed Value (S_t)
1/2024	−213
2/2024	−423.6
3/2024	−190.44
4/2024	−85.176
5/2024	50.5296
6/2024	89.21184
7/2024	−216.315264
8/2024	−302.5261056
9/2024	0.9895578
10/2024	−202.40417688
11/2024	−339.56167075
12/2024	−19.4246683

——- 3 marks

Conclusion:

In this case, **α=0.4** gives a smoother trend with less fluctuation compared to α=0.6. This is because a smaller α places more weight on previous smoothed values, making the model less responsive to recent changes. ------- 1 marks

b) Two judges ranked 8 ice skaters in a competition according to the table below. Compute Spearman's rank correlation coefficient between the ranks given by the two judges. [4 Marks]

Solution:

X:	2	5	3	7	8	1	4	6
Y:	3	2	6	5	7	4	1	8

Rank x	Rank y	$d = x - y$	d^2
2	3	-1	1
5	2	3	9
3	6	-3	9
7	5	2	4
8	7	1	1
1	4	-3	9
4	1	3	9
6	8	-2	4
			46

2 mark

$n = 8$

$$\rho = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6(46)}{8(63)} = 1 - 0.5476$$

$$\rho = 0.4524$$

2 mark

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